

Hypothesis Testing with One Sample

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● Question 1

☑ 0/1 pt ↻ 3 ↺ 19

You are conducting a study to see if the accuracy rate for fingerprint identification is significantly less than 0.53. You use a significance level of $\alpha = 0.10$.

$$H_0: p = 0.53$$

$$H_1: p < 0.53$$

You obtain a sample of size $n = 304$ in which there are 150 successes.

What is the test statistic for this sample? (Report answer accurate to three decimal places.)

test statistic = _____

What is the p-value for this sample? (Report answer accurate to four decimal places.)

p-value = _____

The p-value is...

- ☐ less than (or equal to) α
- ☐ greater than α

This test statistic leads to a decision to...

- ☐ reject the null
- ☐ accept the null
- ☐ fail to reject the null

As such, the final conclusion is that...

- ☐ There is sufficient evidence to warrant rejection of the claim that the accuracy rate for fingerprint identification is less than 0.53.
- ☐ There is not sufficient evidence to warrant rejection of the claim that the accuracy rate for fingerprint identification is less than 0.53.
- ☐ The sample data support the claim that the accuracy rate for fingerprint identification is less than 0.53.
- ☐ There is not sufficient sample evidence to support the claim that the accuracy rate for fingerprint identification is less than 0.53.

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● Question 2

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Test the claim that the proportion of people who own cats is significantly different than 10% at the 0.05 significance level.

The null and alternative hypothesis would be:

$$H_0: p = 0.1 \quad H_0: \mu = 0.1 \quad H_0: \mu \geq 0.1 \quad H_0: p \leq 0.1 \quad H_0: \mu \leq 0.1$$

$$H_1: p \neq 0.1 \quad H_1: \mu \neq 0.1 \quad H_1: \mu < 0.1 \quad H_1: p > 0.1 \quad H_1: \mu > 0.1$$



$$H_0: p \geq 0.1$$

$$H_1: p < 0.1$$



The test is:

left-tailed two-tailed right-tailed



Based on a sample of 100 people, 3% owned cats

The p-value is: _____ (to 2 decimals)

Based on this we:

- ☐ Reject the null hypothesis
- ☐ Fail to reject the null hypothesis

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● Question 3

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The drug OxyContin (oxycodone) is used to treat pain, but it is dangerous because it is addictive and can be lethal. In clinical trials, 272 subjects were treated with OxyContin and 233 of them developed nausea (based on data from Purdue Pharma L.P.). Use a 10% significance level to test the claim that the proportion of OxyContin users that develop nausea is less than 90%.

Procedure:

Assumptions: (select everything that applies)

- ☐ Population standard deviation is unknown
- ☐ Normal population
- ☐ Population standard deviation is known
- ☐ Simple random sample
- ☐ Sample size is greater than 30
- ☐ The number of positive and negative responses are both greater than 10

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● Question 4

0/1 pt 3 19

A college administrator claims that 75% of college students purchase their books from the campus bookstore. You think this is inaccurate and form a random sample of 39 students at that college and find that 22 of them purchased their books from the bookstore. Test the administrator's claim using a level of significance of 10%.

a. What type of test will be used in this problem?

Select an answer

b. What evidence justifies the use of this test? Check all that apply

- ☐ The original population is approximately normal
- ☐ The population standard deviation is not known
- ☐ $np > 5$ and $nq > 5$
- ☐ The population standard deviation is known
- ☐ There are two different samples being compared
- ☐ The sample size is larger than 30
- ☐ The sample standard deviation is not known

c. Enter the null hypothesis for this test.

H_0 ? ?

d. Enter the alternative hypothesis for this test.

H_1 : ? ?

e. Is the original claim located in the null or alternative hypothesis?

Select an answer

f. What is the test statistic for the given statistics?

g. What is the p -value for this test?

h. What is the decision based on the given statistics?

Select an answer

i. What is the correct interpretation of this decision?

Using a _____ % level of significance, there sufficient evidence to the claim that 75% of college students purchase their books from the campus bookstore.

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● Question 5

 0/1 pt  3  19

The US Department of Energy reported that more than 42% of homes were heated by natural gas. A random sample of 246 homes in Oregon found that 120 were heated by natural gas. Test the claim using a 5% significance level.

a. What type of test will be used in this problem?

Select an answer

b. What evidence justifies the use of this test? Check all that apply

- ☐ The population standard deviation is not known
- ☐ The sample size is larger than 30
- ☐ The original population is approximately normal
- ☐ There are two different samples being compared
- ☐ $np > 5$ and $nq > 5$
- ☐ The sample standard deviation is not known
- ☐ The population standard deviation is known

c. Enter the null hypothesis for this test.

H_0 ? ?

d. Enter the alternative hypothesis for this test.

H_1 : ? ?

e. Is the original claim located in the null or alternative hypothesis?

Select an answer

f. What is the test statistic for the given statistics?

g. What is the p -value for this test?

h. What is the decision based on the given statistics?

Select an answer

i. What is the correct interpretation of this decision?

Using a % level of significance, there Select an answer sufficient

evidence to Select an answer ▾ the claim that more than 42% of homes were heated by natural gas.

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● Question 6

☑ 0/1 pt ↻ 3 ↺ 19

Company records indicate that less than 25% of employees commute to work using public transportation. You suspect this is incorrect and for a sample of 118 employees, and find that 19% of them take public transportation to work. Test the claim using a 1% level of significance.

a. What type of test will be used in this problem?

Select an answer

b. What evidence justifies the use of this test? Check all that apply

- ☐ $np > 5$ and $nq > 5$
- ☐ The sample size is larger than 30
- ☐ The population standard deviation is known
- ☐ There are two different samples being compared
- ☐ The sample standard deviation is not known
- ☐ The original population is approximately normal
- ☐ The population standard deviation is not known

c. Enter the null hypothesis for this test.

H_0 : _____

d. Enter the alternative hypothesis for this test.

H_1 : _____

e. Is the original claim located in the null or alternative hypothesis?

Select an answer

f. What is the test statistic for the given statistics?

g. What is the p -value for this test?

h. What is the decision based on the given statistics?

Select an answer

i. What is the correct interpretation of this decision?

Using a _____ % level of significance, there sufficient evidence to the claim that less than 25% of employees commute to work using public transportation.

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● Question 7

 0/1 pt  3  19

An incubation period is a time between when you contract a virus and when your symptoms start. Assume that the population of incubation periods for a novel coronavirus has the population standard deviation of 2.2 day(s). By surveying randomly selected local hospitals, a researcher was able to obtain the following sample of incubation periods of 41 patients:

7	6	9	8	7
6	7	13	12	10
8	11	5	5	6
4	9	11	8	8
6	11	9	8	8
10	11	7	6	9
5	5	12	9	4
9	11	5	9	11
9				

(Note: The average and the standard deviation of the data are respectively 8.15 day(s) and 2.38 day(s).)

At 10% significance level, test the claim that the average incubation period of the novel coronavirus is different from 6 day(s).

Procedure:

Assumptions: (select everything that applies)

- ☐ Simple random sample
- ☐ Population standard deviation is unknown
- ☐ Sample size is greater than 30
- ☐ Population standard deviation is known
- ☐ Normal population
- ☐ The number of positive and negative responses are both greater than 10

Question Help: [Video](#)

● Question 8

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The Federal Bureau of Investigation (FBI) compiles information on robbery and property crimes by type and selected characteristics and publishes its findings in Population-at-Risk Rates and Selected Crime Indicators. According to that document, the mean value lost to purse snatching was 415 dollars in 2004. This year, the mean and standard deviation of the 17 randomly selected purse snatching offenses are 391.69 dollars and 26.61 dollars, respectively. At the 1% significance level, do the data provide sufficient evidence to conclude that this year's mean value lost to purse snatching is different from the mean value lost in 2004? You may also assume that the sample is from a normal population.

Procedure:

Assumptions: (select everything that applies)

- ☐ Population standard deviation is unknown
- ☐ Normal population
- ☐ Sample size is greater than 30
- ☐ The number of positive and negative responses are both greater than 10
- ☐ Simple random sample
- ☐ Population standard deviation is known

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